

PROGRAM – 6

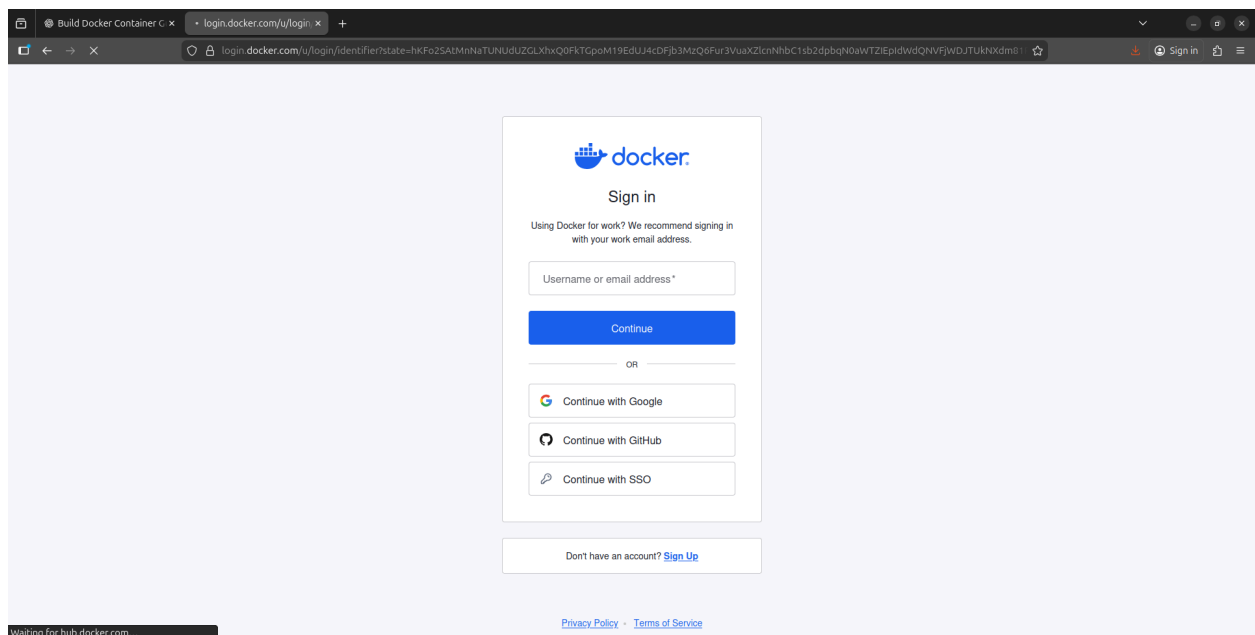
Deploy an Automated Build Pipeline using Docker Hub

Project Name: program_devops_6

Technologies: GitHub, Docker Hub, Docker, Node.js

Step 1: Login to Docker Hub from terminal

`docker login`

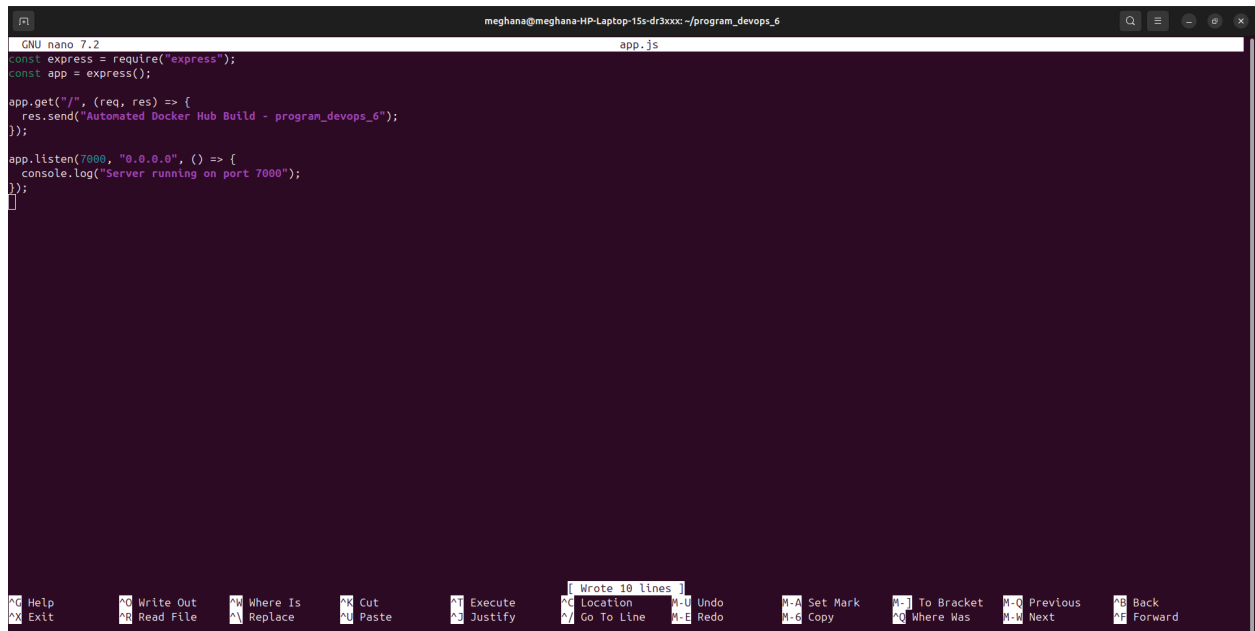


Step 2: Create project directory

```
mkdir program_devops_6 cd program_devops_6
```

Step 3: Create Node.js application

```
nano app.js
```



```
GNU nano 7.2 app.js
const express = require("express");
const app = express();

app.get("/", (req, res) => {
  res.send("Automated Docker Hub Build - program_devops_6");
});

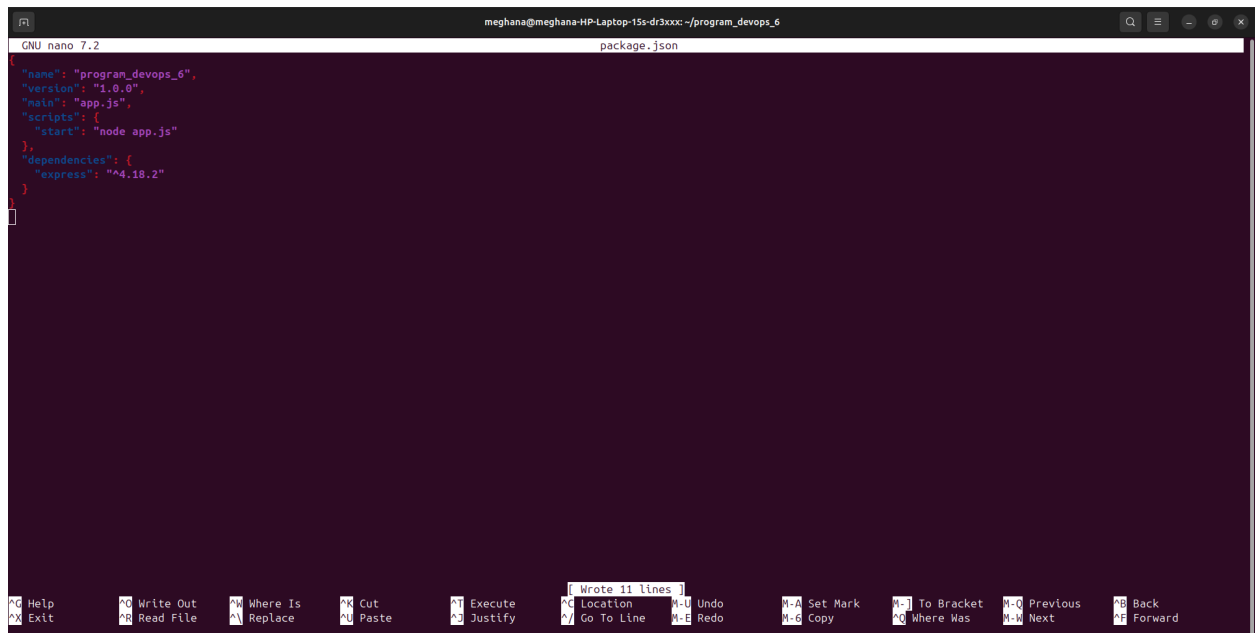
app.listen(7000, "0.0.0.0", () => {
  console.log("Server running on port 7000");
});
```

Wrote 10 lines

Help Exit Write Out Read File Where Is Replace Cut Paste Execute Justify Location Go To Line Undo Redo Set Mark Copy To Bracket Where Was Previous Next Back Forward

Step 4: Create package.json

nano package.json

A screenshot of a terminal window showing the nano text editor editing a file named package.json. The editor's title bar indicates the user is meghana on a laptop, in the directory /program_devops_6. The file content is a JSON object with fields: name (program_devops_6), version (1.0.0), main (app.js), scripts (start: node app.js), and dependencies (express: ^4.18.2). The bottom status bar shows various keyboard shortcuts for nano, and a message indicates 11 lines have been written.

```
GNU nano 7.2 package.json
{
  "name": "program_devops_6",
  "version": "1.0.0",
  "main": "app.js",
  "scripts": {
    "start": "node app.js"
  },
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

Step 5: Create Dockerfile

nano Dockerfile

A screenshot of a terminal window showing the nano text editor editing a file named Dockerfile. The editor's title bar shows the same user and directory as the previous screenshot. The file content includes instructions to set the base image to node:18-alpine, set the working directory to /app, copy package.json and app.js, install npm dependencies, expose port 7000, and set the command to run npm start. The bottom status bar shows keyboard shortcuts and a message indicating 13 lines have been written.

```
GNU nano 7.2 Dockerfile
FROM node:18-alpine

WORKDIR /app

COPY package.json .
RUN npm install

COPY app.js .

EXPOSE 7000

CMD ["npm", "start"]
```

Step 6: Initialize Git & push to GitHub

git init git add . git commit -m "Initial commit for Docker Hub automated build"

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_6
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ mkdir program_devops_6
cd program_devops_6
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ git init
git add .
git commit -m "Initial commit for Docker Hub automated build"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/meghana/program_devops_6/.git/
[master (root-commit) 77997e5] Initial commit for Docker Hub automated build
3 files changed, 34 insertions(+)
create mode 100644 Dockerfile
create mode 100644 app.js
create mode 100644 package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$
```

Create GitHub repo:

program_devops_6

Add remote & push:

git branch -M main git remote add origin

https://github.com/<your-username>/program_devops_6.git git push

-u origin main

```
meghana@meghana-HP-Laptop-15s-dr3xxx: ~/program_devops_6
meghana@meghana-HP-Laptop-15s-dr3xxx:~$ mkdir program_devops_6
cd program_devops_6
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano app.js
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ nano Dockerfile
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ git init
git add .
git commit -m "Initial commit for Docker Hub automated build"
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/meghana/program_devops_6/.git/
[master (root-commit) 77997e5] Initial commit for Docker Hub automated build
3 files changed, 34 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.js
 create mode 100644 package.json
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$ git branch -M main
git remote add origin https://github.com/meghana361/program_devops_6.git
git push -u origin main
Username for 'https://github.com': meghana361
Password for 'https://meghana361@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 726 bytes | 726.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/meghana361/program_devops_6.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
meghana@meghana-HP-Laptop-15s-dr3xxx:~/program_devops_6$
```

Step 7: Create Docker Hub Automated Build (UI)

On Docker Hub:

1. Login → Repositories
2. Click Create Repository
3. Repo name: prog6

Step 8: Connect GitHub to Docker Hub

Step 9: Configure Automated Build

- Open Docker Hub repository program_devops_6
Go to Builds
Enable Automated Builds

Step 10: Trigger Automated Build

```
git add .
```

```
git commit -m "Trigger Docker Hub automated build"
```

```
git push
```

Step 11: Verify Build on Docker Hub

Step 12: Pull Image from Docker Hub

